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Research on the application of artificial intelligence technology in the field of urban security

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Abstract: Artificial intelligence is one of the core factors that determine the future development of the security field. As one of the main means to maintain public security in modern society, security technology is of great significance to ensure the harmonious and stable development of society. In order to strengthen the application of artificial intelligence in the field of security and improve the automation and intelligence level of security technology, it is necessary to continuously study the application scheme of artificial intelligence in the field of security so that the security products can continuously meet the new needs brought by social development. In view of this, this paper mainly discusses the application of artificial intelligence technology in the field of security. First of all, the development level and stage of AI in the field of security are described. Secondly, the types of AI technology applied in the field of security are introduced. Thirdly, the practical application of AI in the field of security is summarized. Finally, it discusses the development prospect and trend of smart security in smart city construction.

1. Introduction

In recent years, the market scale of domestic security industry has maintained stable growth, and its market prospect is promising. With the development of artificial intelligence, intelligence has become the general trend of the industry. Intelligent security has gradually become the direction of security industry transformation and upgrading, accounting for an increasing proportion in the security industry. Artificial intelligence has become an indisputable place for security enterprises. From the perspective of technology application and scene, AI's application in the field of security mainly involves face and vehicle recognition, including biometric recognition technology, big data and video structured technology.[1]

From the perspective of industry application, at present, intelligent security has application scenarios in public security, transportation, buildings, finance, industry, civil and other fields. For example, in the field of public security, it mainly involves the three-tier application of figure detection, actual combat and prediction, as well as its actual needs before, during and after the event. In the field of transportation, in the future, AI technology can be used to analyze urban traffic flow in real time, adjust traffic light interval, shorten vehicle waiting time, etc., so as to reasonably allocate resources to improve the traffic efficiency of urban roads. In the field of intelligent building, AI technology can be used to comprehensively control the security and energy consumption of the building, and realize real-time monitoring of people, vehicles and objects entering and leaving the building to ensure the safety of the

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core area.

Therefore, urban security is the cornerstone of urban development. With the application scenarios of "artificial intelligence (AI) +" becoming more and more diversified, artificial intelligence is a good prescription for the safety problems in urban governance and operation, which helps to create a smart, healthy and safe urban environment. In the report "artificial intelligence helps Cyberspace Security: patterns and practices", artificial intelligence technologies such as computer visualization, biometric identification, and automatic detection of human moving lines have been introduced into the field of urban security monitoring, which can realize "real-time early warning and first-time alarm of criminals who break into their homes" and "identify and prevent potential violent incidents around campus". It provides an important guarantee for the safety of family, campus and large public places.

2. Key technologies of artificial intelligence in security field

With the development of artificial intelligence technology, AI technology is constantly integrated into the application of modern security, so as to promote intelligent security forward to intelligent security, and provide security for the construction of smart city. The development of AI in the field of security is mainly due to the continuous maturity of video structure, biometrics, object recognition and big data technology.

2.1 Video structure

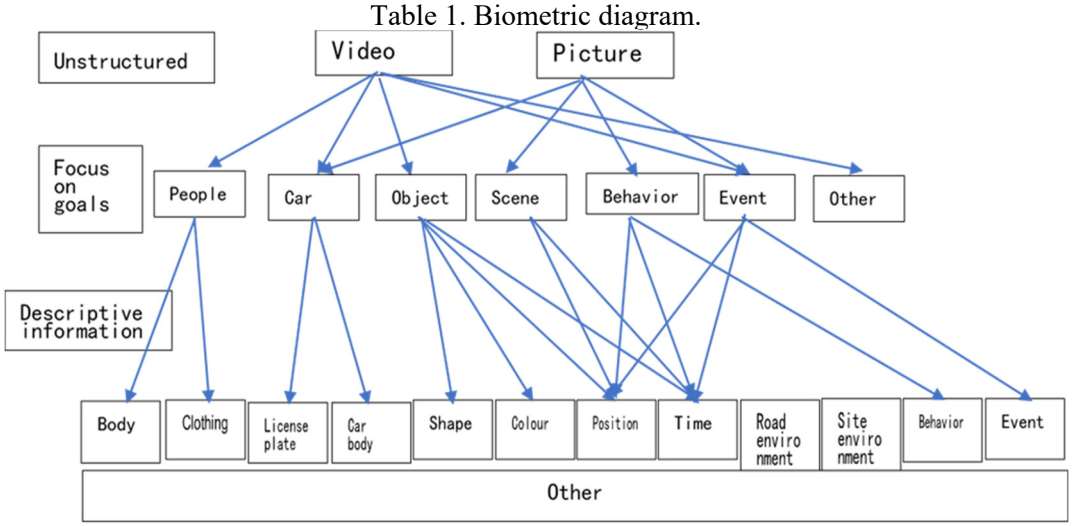
Video structure is the basis and premise of biometrics and object recognition. It can be divided into three steps in the technical field: target detection, target tracking and target attribute extraction. In the process of object detection, we extract the foreground object from the video, and then recognize whether the foreground object is an effective target (people, vehicles, faces, etc.) or an invalid target (leaves, shadows, lights, etc.).[2] In the process of target detection, it is mainly applied to target detection, face detection, vehicle detection and other technologies. The target tracking process is to achieve the continuous tracking of a specific target in the scene, and obtain a high-quality picture as the capture picture of the target from the whole tracking process. In the process of target tracking, it is mainly applied to multi-target tracking, target fusion and target scoring technology.

The target attribute extraction process is to identify the target attributes in the detected target image, and judge which visual features the target has, such as gender, age, dress, vehicle type, color and other attributes of the personnel target. The process of object attribute extraction is mainly based on the feature extraction and classification technology of deep learning network structure. After AI has structured video data, the speed of video search will be greatly improved. In the past, after the occurrence of a case, the police could only search the video manually and check the target personnel one by one, but after the video data is structured, it only takes a few seconds to find a suspect from a million level target database. At the same time, deep mining of structured data can also achieve the prediction function. In addition, structured video data takes up less memory space, which can effectively reduce the pressure of transmission and storage in today's data explosion.

2.2 Biometrics

Biometric technology is a technology that uses the inherent physiological and behavioral characteristics of human body to identify individuals. With the development of science and technology, biometric technology has become an important way of personal identification or authentication technology. As an important branch of biometrics, face recognition can recognize users in the most natural and intuitive non-contact way, which is more easily accepted by users, and is also the most widely used technology in the field of public security.[3]

Table 1. Biometric diagram.



The actual needs of public security are mainly the tracking, analysis and investigation of "people, vehicles and objects". At present, the public security organs have the most urgent technical needs to deal with "people". In the field of public security actual combat, face recognition can not only confirm the identity, but also achieve face comparison, face tracking, live detection, etc., help the police to quickly lock the identity of the target object in the crowd, achieve pre prevention, pre-warning in the event, post verification, greatly improve the efficiency of police handling cases and save police force.

2.3 Object recognition

At present, in the field of security, the typical application of object recognition is vehicle recognition. Through peripheral trigger and video trigger, vehicle image can be collected and license plate can be recognized automatically. The recognition of vehicle shape characteristics and license plate can not only master vehicle attributes, but also master owner attributes, related person attributes, behavior characteristics, etc., and combine with biometrics to build a three-dimensional prevention and control system.

In practical application, vehicle recognition can be compared in series with vehicle attributes, owner attributes and other data to give early warning to high-risk vehicles. The extraction of color, model, shelter plate, hanging parts and other features can effectively solve the problem of "deck car" that puzzles traffic police. Screening passing vehicles at checkpoints can filter trusted vehicles on the basis of establishing a large number of databases, so as to quickly lock the target vehicles and reduce the pressure of police work.

2.4 Big data technology

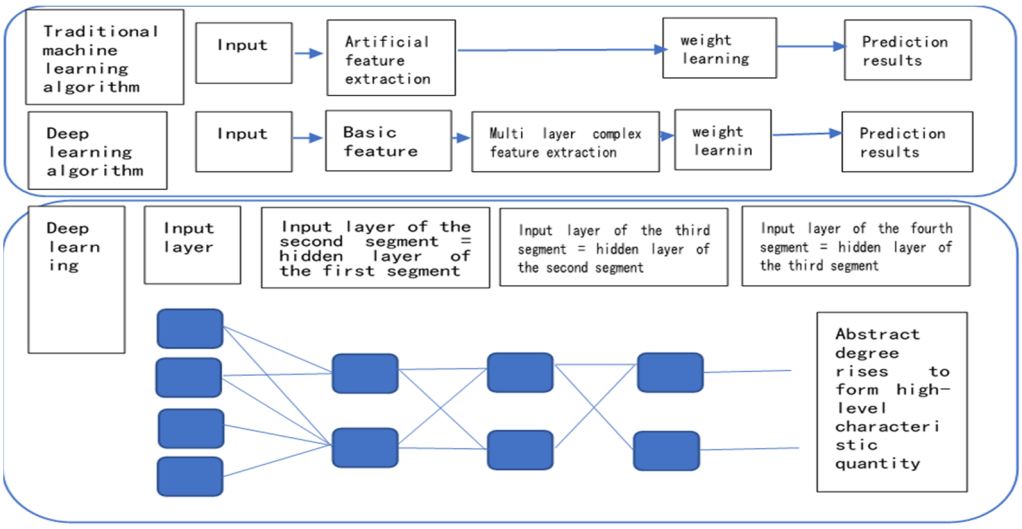
Big data technology can integrate massive unstructured, semi-structured and structured data, and analyze and calculate these data. Big data technology provides powerful distributed computing ability and knowledge base management ability for AI, which is an important support for AI analysis, prediction and self-improvement. [4]It consists of three parts: massive data management, large-scale distributed computing and data mining. Massive data management is used to collect and store the all-round data resources involved in the application of artificial intelligence, and to accumulate data based on the time axis, so as to reflect the law of real things in the time dimension. At the same time, valuable data can be formed through big data mining and analysis, which can provide more effective data samples for deep learning. Big data technology can greatly reduce labor costs, improve efficiency, and even make some tasks that cannot be completed manually possible in some aspects. For example: face, personnel database retrieval, ID card database duplicate personnel search, through semantic description from the video for specific color vehicle search, license plate search, image search, Video Association and other applications.

3.The practical application of artificial intelligence in the field of security

3.1 Application of AI deep learning technology in security field

At present, the AI deep learning architecture network mainly includes self encoder, Boltzmann machine and convolutional neural network, among which the deep convolutional neural network based on convolutional neural network (DCNN) has become the research hotspot of speech and image recognition technology. Because it is similar to biological neural network, has less weight number and lower model complexity. The deep convolution neural network mainly includes three layers: first, the input layer. The data collected by the device is then imported into the system. Second, the hidden layer. The hidden layer is divided into two layers: high layer and low layer. The high layer is the full connection layer, and the low layer is composed of the convolution layer and the lower sampling layer alternately in pairs. Third, the output layer. The output layer can use acquisition logic regression, SoftMax regression or support vector machine to classify images.

Table 2. Schematic diagram of deep convolution neural network.



According to the theory of "local perceptual field", the local adjacent pixels are closely related, while the pixels far away are weakly related. Therefore, only observing the adjacent pixels, and synthesizing the information of local observation at the upper level, can grasp the overall situation. With the method of local sensing area 10 × 10, 1012 full connection parameters between each layer of 1000 × 1000 pixel image can be reduced to 108 local connection parameters. By using the method of weight sharing, the parameters of 108 local connections in each image layer can be reduced to 100. Convolution neural network is the neural network that realizes "local perceptual field" and "weight sharing" by convolution. Based on the AI deep learning technology of deep convolution network, the monitoring equipment can automatically identify and judge the obtained monitoring image, understand the video content by itself, calculate a large number of potential information in the scene, and realize intelligent judgment, such as case law analysis, financial fraud analysis, public transport optimization.

3.2 Application of AI video structure technology in security field

On the basis of AI deep learning technology, the intelligent video analysis technology formed by establishing supporting technical measures is AI video structure technology. There are three steps: first, target detection. The process of extracting foreground objects from image information and identifying them is target detection. Second, target tracking. Target tracking is a kind of intelligent judgment process, which is to lock and shoot specific foreground targets and capture some key actions. Third, target attribute extraction. The process of recognizing, extracting, classifying and storing the visual features of specific foreground objects is object attribute extraction. The above three steps integrate AI deep learning, machine vision, data processing and other technologies to realize the automatic retrieval,

judgment and response functions of security technology.

3.3 Application of AI big data technology in security field

AI big data technology is a technology that uses artificial intelligence to screen, calculate and mine the potential value of massive data. It can greatly enhance the accuracy of artificial intelligence in the field of security and provide a variety of security functions. So we can develop more security products suitable for various security scenarios, such as public security management system, smart home security solutions, smart police security system.

4.The key link of the practical application of artificial intelligence in the field of security

From the perspective of scenarios, the practical application of intelligent security can be analyzed according to the time of the event, the type of event, how to prevent the event and how to replay the event process.

4.1 The effect of starlight level can ensure that you can see clearly in 24 hours

Most of the events happen at night, while ordinary products are black and white with infrared light. These videos will lose a lot of color information and key details when viewing events. So we must achieve 24-hour full color in effect. Taking the warning series of a company as an example, all of them adopt star light and super star light level sensors. In the dark or low light monitoring environment, there is no need for external light compensation equipment, which can also restore the perfect color image.

4.2 Warning function: intelligent early warning, automatic tracking and early prevention

The essence of security is prevention. How to effectively prevent the occurrence of events, or how to eliminate the events in the bud, is the just need of intelligent security. When someone or a vehicle intrudes into the warning area, if a variety of natural on-site interferences can be filtered out through the built-in intrusion detection operator, the acousto-optic early warning system will be triggered after the accurate identification of the intrusion target, the built-in white light will start flashing, and the voice prompt will be carried out synchronously to frighten the intruder or a vehicle, at the same time, the linkage backend will alarm and push.

Taking the warning ball as an example, when someone intrudes into the warning area, the product can immediately synchronize the timing of the intruder or vehicle. When the target's retention time in the limited area exceeds the preset time (the time can be set, the default is 10 seconds), the warning ball starts to raise the warning level, starts the tracking linkage and laser scanning system, and the warning ball will be turned on Laser beam invades people or vehicles, sends out alarm sound and tracks the ball machine.

4.3 Intelligent deterrence.

A complete intelligent security product must have a sound and light deterrent system to ensure that the system can actively take deterrent strategies to prevent the occurrence of events in the event of prediction. Taking the warning ball as an example, the full spectrum fill light is realized. Through the application of the full spectrum fill light system, the built-in white light is used to fill the light for the short-distance scene, and the built-in infrared light is used to fill the light for the long-distance scene, so that the imaging effect at night continues to be improved. The alert system integrates the sound prompt, white light flashing, laser tracking and other linkage early warning modes. The biggest difference between the alert system and the traditional security system is the intelligence, because the intelligent security system can realize the intelligent judgment through the machine, so as to realize the things that people want to do as much as possible. The traditional security system is highly dependent on people, which is very labor-intensive. The landing of AI must be combined with the application scenarios. The most fundamental change is to make retroactivity into pre-warning, which will make the popularization of intelligence inevitable. It is necessary to liberate human resources and more efficient security technology to build a smart city.

4.4 Warning function

It can intelligently analyze the area perimeter and personnel behavior, and flexibly deal with all kinds of intrusion, regional wandering, regional staying and so on. According to different levels of alarm situation, four levels of early warning and user-defined alarm mode can reduce the energy of backstage watchmen and reduce the cost of manpower and time. In the era of intelligence, the power of subversion is obvious.

5. Development prospect and trend of smart security in smart city construction

With the application of AI, big data and other technologies in the field of security. The future development trend of smart security is mainly shown in three aspects.

5.1 Combination of "cloud + end" full intelligent perception AI and front-end perception equipment

By giving the front-end edge computing ability, some intelligent analysis functions such as human image recognition, vehicle recognition and behavior recognition can be brought to the front to realize the "cloud + end" AI security overall architecture. It not only meets the intelligent application requirements of urban security for multiple subdivision scenes, but also reduces the pressure of data transmission on the network and the dependence on the data center through calculation and data marginalization, so as to improve the intelligent efficiency of the whole network.

5.2 Man machine cooperation and intelligent operation

In the 2030 AI life report released by Stanford University in the United States, the development of artificial intelligence is comprehensively evaluated, and the intelligent collaborative system of man-machine mutual compensation and enhancement is listed as one of the important development trends of AI in the future. Human machine collaboration needs machine cognitive ability based on human state model, human machine knowledge sharing ability based on knowledge map and global planning ability based on intelligent reasoning. [5]In the field of intelligent security, human-machine system can play a role in remote emergency command, joint search and rescue at the scene of an accident, operation in areas inaccessible to human beings, and realize the active cognition and cooperation of human-machine organizations. For example, in a critical environment, citizens can send help signals to the video monitoring system through gestures.

5.3 Global cognitive intelligence

Security big data is a collection of data from multiple departments and systems at different time points. The data value density is very low, and the lack of global data analysis ability seriously restricts the application of security data. In the future, it is the key direction of intelligent security development to let machine learning process large-scale, full-scale and multi-source data that cannot be understood by human beings in real time, to understand the complex hidden laws that human beings have not discovered from the massive data, and finally to formulate the optimal strategy beyond the local suboptimal decision-making of human beings from a global perspective.

In practice, the exploration of Shanghai is worth learning. In recent years, Shanghai has comprehensively promoted the construction of smart city facing the future, and proposed the goals of strengthening intelligent perception, optimizing the comprehensive operation of the city, strengthening decision-making assistance, and promoting the quality and efficiency of public services. Artificial intelligence has gradually become a new trend of urban management. In 2018, the Shanghai public security and traffic police department installed the "intelligent traffic light" system in the original Puxi Expo site, making the traffic lights "smart". The system can realize traffic queue length detection, signal light fault detection, traffic illegal behavior capture, behavior and event detection and push, etc. It can also use "brain" to analyze and judge, and give better scheme of signal control and traffic organization. Relying on high and new technology, Shanghai has realized the "e-police" supervision of illegal behaviors of motor vehicles such as "illegal horn", "not wearing safety belt", "driving with mobile phone", "not following the provisions of alternative traffic", "not avoiding pedestrians according to the

regulations" and "pedestrians running red lights", etc., and the supervision information is fed back to motor vehicle drivers by SMS at the speed of seconds. Relevant departments in Shanghai will also share GPS dynamic monitoring data of key vehicles such as long-distance passenger transport, hazardous chemicals transportation and residue transportation with relevant enterprises, and use the real-time operation data of vehicles to investigate and deal with traffic violations such as speeding, illegal operation in the early hours of the morning and breaking into the forbidden zone.

6.Conclusion

The development level of artificial intelligence in the field of security is reversed. The first level is the intelligent recognition technology of video surveillance terminal, such as face recognition, license plate number recognition. The second level is the intelligent analysis technology of security data, which can automatically store data and conduct data essence analysis. The third level is big data analysis and operation technology, which can simulate and analyze the form of environmental security for a period of time, carry out in-depth analysis and trend prediction. There are three stages in the development of artificial intelligence in the security industry. The first stage is the application of artificial intelligence technology for basic security products, such as monitoring image recognition, voice recognition, access control system. The second stage is the application of advanced artificial intelligence. Security products can simulate human thinking, replace part of human work, and can cross platform, cross equipment networking and cooperation. The third stage is the application of super artificial intelligence. In some areas with sufficient technology reserves, the performance of artificial intelligence can be better than that of human beings, which is the top stage of the development of artificial intelligence technology.

Now, intelligent security has entered the era of big data and artificial intelligence. Artificial intelligence based on machine vision and deep learning technology has been widely used in public security control, traffic management, criminal investigation and other business scenarios. In the environment without human intervention, the computer can automatically analyze the content of the camera shooting, including target detection, target segmentation extraction, target recognition, target annotation, target tracking. Like blood, security data permeate all aspects of urban security, transportation, community, education, production, etc. Dealing with massive multi-source heterogeneous data is a problem that intelligent security must face. At present, the public security system database accumulates a large number of vehicles, personnel, social relations and other information, as well as a large number of high-risk personnel, high-risk vehicle information. At the same time, city cameras, intelligent mobile terminals, sensors are producing a large number of social security data at all times, and the city security information is hidden in these data. Through the multi-dimensional collision analysis, combined with the actual combat business process, the deep-seated value of data is continuously explored, and a "multi-dimensional intelligent perception prevention and control network" is constructed to break through the data barriers and serve the application of the whole police and all government departments. Multi dimensional big data system can collect and associate personnel data, vehicle data and IOT data. Through the IOT information service with video as the core, it collides with public security business data and government social data, so as to achieve the functions of integrated retrieval, whole network collision, relationship tracking, track completion, track prediction, etc.

Artificial intelligence is the future of the security field. On the road to the future, there are many obstacles and difficulties to overcome, but the overall trend is optimistic. We firmly believe that only with autonomous, personalized and constantly evolving artificial intelligence brain, can we solve the increasing demand in the security field. The deep integration of virtual and reality is just around the corner. From smart industry and smart city to smart country and smart world, more and more open and vast world is waiting for us to explore.

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